



**Verified Carbon
Standard**

BLUE FLAME FOR COMMERCIAL KITCHENS IN GHANA



Document Prepared by



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The proposed project "Blue Flame for Commercial Kitchens in Ghana" is a voluntary initiative by Envirofit International Inc. to improve access to clean and efficient cooking solutions, for commercial cooking in Ghana. The project involves the dissemination of energy and emission efficient liquefied petroleum gas (LPG) stoves / burners (hereinafter referred to as 'LPGS') to commercial kitchens operating in small and medium-sized enterprises, schools or other commercial applications (SMEs), across multiple locations in Ghana.

The project aims to replace traditional cooking fuel and technology, primarily three-stone fires or inefficient biomass stoves using non-renewable biomass fuel such as firewood and/or charcoal—with LPGS. The project LPG based / burners. The project devices have high thermal efficiency ($\geq 40\%$) and significantly reduce greenhouse gas (GHG) emissions and indoor air pollution (IAP) attributed to use of traditional biomass cooking practices. The initiative will help mitigate deforestation and health hazards caused by the widespread use of traditional and inefficient non-renewable biomass (NRB) on traditional / inefficient stoves, which is a major driver of land degradation and respiratory illnesses in the region.

The project is designed to benefit stakeholders from carbon finance mechanisms. Carbon revenues will be used to lower the upfront cost barrier and promote broader uptake of LPGS, enhancing clean fuel access in Ghana. The project's management system will record LPGS' unique serial numbers, geotagged distribution data, and end-user information to ensure traceability and transparency of project devices while avoiding double counting.

In the absence of the project, the baseline scenario would have been the continued reliance on inefficient, traditional modes of cooking (3 stone/open fires, traditional stoves) or high-emission fossil fuels. By transitioning to LPGS, the project displaces the use of carbon-intensive fuels, leading to emission reductions.

Over the first seven-year renewable crediting period, the project is expected to achieve 2,100,000 tCO₂e in emission reductions, with an annual average of 300,000 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	15-April 2024 – present	VCS	TBD	-

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	03 – Energy demand	
Project activity type	Type II – Energy efficiency improvement projects	

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Eligibility

1.4.1 General eligibility

Project eligibility under the scope of the VCS program:

Scope	Eligibility Justification
The Seven Kyoto Protocol GHGs	The project boundary includes Kyoto GHGs viz. CO ₂ , CH ₄ and N ₂ O in the baseline. Thus, the project is applicable to this scope.
Ozone-depleting substances (ODS)	NA. The project does not involve ODSs.
Project activities supported by a methodology approved under the VCS program through the methodology development and review approval process	The project uses the methodology “VM0050 “Energy Efficiency and Fuel-Switch Measures in Cookstoves”, Version 1.0.” which is approved under the VCS program through the methodology development and review process.
Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded:	Not applicable. The project does not use a methodology approved under other approved GHG program.
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements	Not applicable. The project does not involve REDD+.
The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction	Not applicable. The project replaces non-renewable biomass-fired cookstoves with efficient LPGs, which are classified as clean technology. LPGs offer superior heat transfer and more complete combustion compared to traditional biomass cookstoves. Unlike non-renewable biomass fuels (e.g., wood, charcoal), LPG is relatively cleaner fuel, resulting in significantly lower emissions of smoke, black soot, particulate matter (PM), and greenhouse gases (GHGs). Thus, the project does not generate emissions for the purpose of their reduction later. Without this project, commercial kitchens would have continued to rely on inefficient biomass based cooking, leading to significant GHG emissions.

Additionally, the project is not included in Table 1, Section 2.1.3 of the VCS Standard v4.7 thus complies with the requirements of Section 2.1.3 of the VCS Standard v4.7.

Requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline:

Reference to VCS Standard version 4.7	Requirement	Justification
3.8.1	Non-AFOLU projects shall complete validation within two years of the project start date. Additional time is granted for non-AFOLU projects to complete validation	The project shall complete validation within two years of the project start date.

	where they are applying a new VCS methodology. Specifically, projects using a new VCS methodology and completing validation within two years of the approval of the methodology by Verra may complete validation within four years of the project start date.	
4.1.5	The validation/verification body shall ensure that the project is listed on the project pipeline with a status of under validation before the opening meeting with the project proponent, such opening meeting representing the beginning of the validation process.	The PP shall ensure that the opening meeting, representing the beginning of the validation process, is conducted once the project is listed on the project pipeline with a status of under validation.

Eligibility of applied methodology under the VCS Program:

The applied methodology, VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0, is an approved methodology under the VCS Program thus the applied methodology is eligible under the VCS Program.

The applied methodology does not prescribe any scale or capacity limits for the project. Therefore, the requirements to demonstrate that the project is not a fragmented component of a larger project that would otherwise exceed such limits, or that no single cluster of project activity instances (PAI)s exceeds the capacity limit, are not applicable. For reference, a PAI refers to a unit of LPGS (as per VCS program definitions, v4.5, dated 16 April 2024² and VCS Validation and Verification Manual, v3.2 dated 19 Oct 2016, section 3.2.10, page 26³).

Compliance with other general requirements is described below:

Reference to VCS Standard version 4.7	Requirement	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	The project meets all applicable rules and requirements as set out under the VCS Program and is guided by the principles set out in section 2.2.1.
3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1.	The project uses VCS approved methodology VM0050, version 1.0 along with relevant tools or modules as applicable, ensuring full compliance.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The project applies the latest version 1.0 of VCS approved methodology VM0050. Refer to section 3 of the PD below for more details.

² <https://verra.org/wp-content/uploads/2024/04/VCS-Program-Definitions-v4.5-FINAL-4.15.24.pdf>

³ <https://verra.org/wp-content/uploads/2024/07/VCS Validation Verification Manual v3.2.pdf>

3.1.4	Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	The project and its project activity instances shall remain in compliance with the applicable local laws.
3.1.5	Where projects apply methodologies that permit the project proponent its own choice of model, such model shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or removals).	Not applicable, as the applied methodology, VM0050 (Version 1.0), does not allow PPs to select their own choice of model for quantification of GHG emission reduction.
3.1.6	Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements.	The project uses the methodology “VM0050 v1.0.” which is approved under the VCS program through the methodology development and review process. Thus, it meets all the requirements set out in the VCS program document VCS methodology requirements. The default factors applied if any, in the project are as governed by the applied methodology.
3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	The project shall adhere to the rules and requirements of the VCS Program. While the project may also utilize the rules and regulations of approved GHG programs but in cases where there is a discrepancy or conflict between the rules and requirements of an approved GHG program and those of the VCS Program, the VCS Program rules and requirements shall take precedence.
3.1.8	Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	Not applicable, as applied methodology is approved under VCS program.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 of VCS standard ver 4.7 and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	The project adheres to the latest VCS Program rules, and the effective dates of these requirements set out in Appendix 3 of VCS standard ver 4.7.

1.4.2 AFOLU project eligibility

Not applicable, as the project involves the dissemination of energy efficient LPG stoves (hereinafter referred as LPGS), for commercial kitchens of Ghana.

1.4.3 Transfer project eligibility

Not applicable, as the project is seeking registration only under the Verra registry. The project has not been listed under any other carbon registries.

1.5 Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

Not applicable, as the proposed large-scale project involves distribution of multiple project activity instances (PAIs) in the geographical boundary of Ghana. However, is not being developed as a grouped project. All PAI (project LPGS) shall remain limited to the geographical boundary of Ghana.

1.6 Project Proponent

Organization name	Envirofit International Inc.
Contact person	Nathan Lorenz
Title	Chief Technical Officer
Address	19 Old town square, Suite 238, Fort Collins, Colorado 80524 USA
Telephone	-
Email	nathan.lorenz@envirofit.org

1.7 Other Entities Involved in the Project

Organization name	Climate Secure
Role in the project	Entity Completing the Project Description
Contact person	Rohit Lohia
Title	Director
Address	65, Pragati Apartments, Paschim Vihar, Delhi 110063 India
Telephone	+91 9717093552
Email	info@climate-secure.com

1.8 Ownership

For each PAI, the ownership of the instance shall be clearly demonstrated by virtue of a Declaration / agreement with end user transferring the ownership rights of the carbon credits generated from the project LPGS to Envirofit.

Thus, all VCUs generated from the project activity instances shall be owned by Envirofit International Inc., unless otherwise stated.

1.9 Project Start Date

Project start date	15-March-2024
Justification	Date for distribution of 1st LPG stove under the project

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	15-March-2024 to 14-March-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
15-March-2024 to 31-December-2024	237,500
01-January-2025 to 31-December-2025	300,000
01-January-2026 to 31-December-2026	300,000
01-January-2027 to 31-December-2027	300,000
01-January-2028 to 31-December-2028	300,000
01-January-2029 to 31-December-2029	300,000
01-January-2030 to 31-December-2030	300,000
01-January-2031 to 14-March-2031	62,500
Total estimated ERRs during the first or fixed crediting period	2,100,000
Total number of years	7

Average annual ERRs	300000
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1.12 Description of the Project Activity

The project involves the dissemination of LPGS, in commercial kitchens in Ghana.

Sub-Saharan Africa faces a severe and growing deficit in clean cooking access. As per The Energy Progress Report 2023⁴, despite global efforts, the region continues to lag, with the number of people lacking access to clean cooking, increasing by almost 20 million annually. Gains in clean cooking access have failed to keep pace with population growth, resulting in the number of people without access to clean cooking, getting more than doubled since 1990 and increasing by 60% since 2000⁴. As of 2021, around 900 million people in Sub-Saharan Africa lacked access to clean cooking solutions, and if current trends persist, this figure is expected to reach 1.1 billion by 2030⁴. By 2050, the region is projected to account for 6 out of 10 people globally still without access, with little or no improvement expected⁴.

Ghana has made notable progress in expanding clean cooking access, but significant challenges remain. Between 2017 and 2021, Ghana was among the 20 countries with the largest access deficits, with an estimated 22 million people still lacking access to clean cooking solutions. According to the Demographic and Health Survey 2022⁵, published in January 2024, only 22.4% of the Ghanaian population used clean fuels and technologies for cooking, primarily LPG/natural gas stoves (21.5%). However, 76.2% of the population still rely on solid fuels, with wood (46.5%) and charcoal (29.5%) being the primary sources. Among households using solid fuels, the most common technologies were three-stone stoves/open fires (35%) and traditional solid fuel stoves without chimneys (22%)⁵. Thus, cooking using solid biomass fuel over traditional inefficient stoves is the prominent cooking practice.

The use of unsustainably sourced, non-renewable biomass (NRB) in Ghana is a major driver of deforestation. The use of solid biomass fuels (e.g., wood, charcoal) for cooking releases large amounts of particulate matter (PM), creating hazardous levels of indoor air pollution (IAP). Particulate matter and other pollutants in household air pollution inflame the airways and lungs, impair immune response and reduce the oxygen-carrying capacity of the blood. Each year, 3.2⁶ million people die prematurely from illnesses attributable to the household air pollution caused by the incomplete combustion of solid fuels and kerosene used for cooking. According to estimates by the World Health Organization (WHO), indoor air pollution from household cooking using solid fuels contributes to over 18,000 premature deaths annually in Ghana⁷, with women and children disproportionately affected due to their prolonged exposure while cooking or staying indoors. In developing countries with high mortality rates overall, indoor air pollution ranks fourth in terms of the risk factors that contribute to disease and death (World Health Organization)⁸. Reliance on polluting fuels and technologies also requires significant time for cooking on an inefficient device and gathering and preparing fuel.

⁴ https://trackingsdg7.esmap.org/data/files/download-documents/sdg7-report2023-full_report.pdf

⁵ <https://dhsprogram.com/pubs/pdf/FR387/FR387.pdf>

⁶ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

⁷ <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/household-air-pollution-attributable-deaths>

⁸ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

Although Ghana's annual increase in clean cooking access has been around 1.5% points, this rate is insufficient to achieve universal access by 2030⁴. Even if this rate continues over the next 25 years, it will result in an additional 37.5% points ($=25 \times 1.5\%$) of access, bringing total access to approximately 59.9% by 2050 ($=22.4\% + 37.5\%$). This progress, while better than many other Sub-Saharan African countries, is still inadequate to achieve universal access by 2050. Without significantly scaling up interventions and investments in clean cooking technologies such as LPG, Ghana's current trajectory suggests that a significant portion of the population, would still rely on polluting cooking fuels and as per The Energy Progress Report 2023, Ghana is projected to remain among the 45 out of 47 Sub-Saharan African countries that will still lack universal clean cooking access by 2050⁴.

No such study is available on clean cooking access in the commercial kitchens in Ghana, however the practice is not expected to be any different from that represented by The Energy Progress Report 2023.

The project aims to contribute towards improving clean cooking access in Ghana by distributing energy-efficient LPG stoves to commercial kitchens across the country. This initiative is one of the efforts to promote the adoption of clean fuels and technologies, reducing reliance on traditional biomass and improving health and environmental outcomes.

Technology Measure:

LPGS fall under the category of clean fuels and technology⁹. They offer superior heat transfer and significantly improved fuel combustion compared to traditional or inefficient biomass stoves. This results in reduced smoke, black soot, and particulate matter (PM) emissions. Compared to solid biomass cookstoves, LPG reduces harmful emissions by cutting PM2.5 and carbon monoxide (CO) emissions by more than 90%¹⁰. Additionally, LPG use helps lower overall GHG emissions by reducing Non-CO₂ emissions from incomplete biomass combustion and mitigating deforestation. Thus, by providing access to clean cooking solutions, LPG stoves not only reduce GHG emissions but also improve indoor air quality (IAP) for users.

The project includes dissemination of the following LPGS, besides others. Specification of these example LPGS is given below:

Stove Type	Manufacturer	Thermal Efficiency	Estimate Life	Fixed / portable
LPG	Envirofit International	50-60%	Up to 7 Year	portable

The project provides thermal energy for cooking in commercial kitchens across Ghana through the distribution of energy-efficient LPGS. The project LPGS is a distributed standalone device, and it has no relation / dependence with other manufacturing/production and systems outside the project boundary. The type of service provided by both the baseline and project cooking device is thermal energy for cooking in commercial kitchens in Ghana.

In the baseline scenario, the thermal energy for cooking in commercial kitchens, were provided using non-renewable biomass-based, traditional inefficient or more carbon intensive fossil fuel stoves. Due to the use of inefficient cooking, the baseline technologies delivered equivalent level of useful energy, with significantly greater pressure on forest resources, higher GHG emissions, and increased exposure to IAP in the baseline.

⁹ Refer footnote 8 on page 16 of the energy progress report 2023

¹⁰ <https://www.sciencedirect.com/science/article/abs/pii/S0048969720372296?via%3Dihub>

The project improves service delivery by offering a cleaner, more efficient alternative that eliminates the use of non-renewable biomass thus reducing GHG emissions, improving indoor air quality, and supporting climate and health co-benefits, while continuing to meet the thermal energy needs of commercial kitchens.

Implementation schedule:

Year	Cumulative # of Project devices (PAIs)
2024-25	500
2025-26	1000
2026-27	1500
2027-28	2500
2028-29	3500
2029-30	5500
2030-30	7500

Management Plan and Operational Framework:

The project will be implemented according to the following plan:

1. The PP is responsible for general management of the project, including appointing all personnel / groups of personnel involved in the project development and implementation.
2. The distribution of the LPGS will be managed via PP's own distribution network or with local distribution partners.
3. The PP will communicate to the project beneficiaries during distribution that PP retains exclusive and unconditional rights to the carbon credit generated by use of project LPGS.
4. The PP will maintain and update the LPGS distribution database continuously. The required data collection includes, but is not limited to, the following.
 - (a) Date of distribution
 - (b) Geographic area of distribution
 - (c) Model/type of project LPGS distributed.
 - (d) Quantity of project LPGS distributed.
 - (e) Name of the beneficiary, address, and telephone number (if available)
 - (f) Unique serial number of project LPGS
 - (g) Primary Baseline fuel / technology being used.
5. The information collected will be stored in the distribution database will serve as the project database for ex-post project sampling and monitoring purposes. Also, this shall serve to avoid any double counting and duplicity of data.
6. The PP will coordinate ex-post monitoring activities as follows:
 - a) Implement the monitoring plan,
 - b) Determine the sample size as per sampling plan and identify the samples to be monitored

- c) Ex-post monitoring - the data will be collected using survey questionnaires designed by experts. The surveys will be carried out by external parties or by project implementer teams experience / trained to conduct such surveys. PP will be responsible for ensuring that staff responsible for monitoring and reporting have received adequate training, as applicable.
- d) Ensure the quality of monitoring data (QA/QC). The monitoring results and statistical robustness will be reviewed internally and approved by competent staff.
- e) Use the data collected for emissions reduction calculations.

The aforesaid plan shall ensure that the project is managed and operated in a standard manner.

1.13 Project Location

Host Country: Republic of Ghana

Region/State/Province etc.: All regions of Republic of Ghana

City/Town/Community etc.: All cities and towns in Republic of Ghana

Physical/Geographical location:

The geographical coordinates of Ghana are approximately 4°44'N to 11°11'N latitude and 1°11'E to 3°15'W longitude¹¹.

The map of Republic of Ghana, for reference purpose, is as follows:



¹¹ https://curriculumresources.edu.gh/wp-content/uploads/2024/10/LM-Geography_Section-5-1.pdf?utm_source=chatgpt.com

The boundary for the Project in terms of a geographical area is defined as the territorial boundary of Ghana. The boundary for each PAI corresponds to the physical location where the distributed LPGS units are used within Ghana.

Additionally, each PAI distributed under the project will include recording of geographical data (e.g., latitude and longitude and/or address), which can be verified through monitoring surveys to ensure accurate tracking and compliance.

1.14 Conditions Prior to Project Initiation

As specified in Section 1.11, access to clean fuel and technology in Ghana is very limited, with the prevalent use of non-renewable biomass (e.g., wood, charcoal) on inefficient cookstoves. The current rate of access to clean fuel and technology is still inadequate to achieve universal access by 2050. Ghana's current trajectory suggests that a significant portion of the population will continue to rely on polluting cooking fuels. According to The Energy Progress Report 2023, Ghana is projected to remain among the 45 out of 47 Sub-Saharan African countries that will still lack universal clean cooking access by 2050. Hence, conditions prior to project initiation involved the continued use of non-renewable biomass fuel for cooking by the target population to meet equivalent thermal energy needs as provided by project LPG stoves.

The project replaces non-renewable biomass-fired cookstoves with efficient LPG stoves, classified as clean fuels and technology. LPG stoves offer superior heat transfer and more complete combustion compared to traditional biomass cookstoves. Unlike non-renewable biomass fuels, LPG is a cleaner fuel, resulting in significantly lower emissions of smoke, black soot, particulate matter (PM), and greenhouse gases (GHGs). Thus, the project does not generate emissions for the purpose of later reduction; rather, it addresses the continued baseline emissions that would persist in the absence of intervention. Without this project, commercial kitchens would have continued to rely on inefficient non-renewable biomass combustion, leading to ongoing high emissions.

The baseline scenario is the same as the condition existing prior to the project.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of LPG stove in Ghana.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

If yes to all:

1.18 Sustainable Development Contributions

SDG	Project Contributions - Relevant SDG Target	SDG Impact Indicator
	Justification	

3 - Good Health and Well Being	Social Sustainability: The project's LPGS will enhance the combustion efficiency of solid biomass fuels, significantly reducing indoor pollutant emissions caused by inefficient burning. This will lead to lower smoke and particulate matter (PM) emissions, minimizing household exposure to harmful pollutants associated with traditional cooking methods.	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	Project Specific Indicator / parameter: % users reporting reduction in smoke/PM after shifting to LPGS in project.
4 - Quality Education	Social Sustainability- The distribution of LPGS involves training of personnel and their capacity building for several monitoring tasks e.g., conducting surveys for the monitoring of SDG impacts parameters, water boiling tests etc. The project thus directly contributes to the skill development training.	4.4 By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship	4.4.1 Number of employees provided skill development training. Monitoring parameter: Number of people receiving formal and/or non-formal education and training.
7 - Affordable and Clean Energy	Environmental Sustainability: The project by distributing LPGS will provide access to clean cooking with primary reliance on clean fuels and technology. The project LPGS will be made available at a subsidized cost to beneficiary households. The project will lead to project households getting access to clean cookstoves (LPGS) by the end of its lifetime.	7.1 By 2030, ensure universal access to affordable, reliable, and modern energy services.	7.1.1 Number of beneficiary establishments. 7.1.2 Proportion of population with primary reliance on clean fuels and technology. Monitoring parameter: 1. Number of Beneficiaries under the project i.e., Number of LPGS distributed under the project. 2. % Users reporting operational project LPGS.
8 - Decent Work and Economic Growth	Economic Sustainability: The project results in generating employment for marketing / sales and distribution / technical employees.	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities 8.5.2 Unemployment rate, by sex, age and persons with disabilities Monitoring parameter: Number of male / female employment created by project.

13 - Climate Action	Environmental Sustainability: The Project is directly related to the target as it contributes towards avoidance of GHG emissions from replacement of non-renewable biomass-based cooking.	13.2 Integrate climate change measures into national policies, strategies, and planning.	13.2.2 Amount of CO ₂ e emissions avoided or sequestered. Monitoring parameter: Total greenhouse gas emissions per year
15 - Life on Land	Environmental Sustainability: The project will reduce reliance on non-renewable biomass from forests, contributing to the conservation of existing forest resources and the protection of natural ecosystems and wildlife habitats	15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.	15.2.1 Progress towards sustainable forest management. Monitoring parameter: Total non-renewable biomass fuel saved after shifting to project device.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable, as default adjustment factor of 0.95 is applied to the GHG emission reduction to account for the leakage emissions as per para 8.3(1) of the applied methodology.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The purpose of the project is to replace the use of non-renewable biomass, primarily on traditional/inefficient technology, for cooking with energy-efficient LPG stoves. Stakeholders were identified through geographic and socioeconomic assessments, community engagement activities, consultation with local authorities, and interaction with NGOs and government representatives. Identified stakeholders include local communities (beneficiaries), NGOs, environmental organizations, academic institutions, government bodies, and project developers.

Legal or customary tenure/access rights	Not applicable. The project focuses on the voluntary distribution of LPG cookstoves to individual commercial kitchens. It does not involve land acquisition, resource extraction, or activities that impact legal or customary tenure rights. No collective or conflicting rights to territories or resources are held or affected by Indigenous Peoples, local communities, or customary rights holders.
Stakeholder diversity and changes over time	The stakeholders encompass local communities, NGOs, government representatives, academia, and project developers, representing diverse social, economic, and cultural backgrounds. Local communities, primarily comprising small and medium commercial cooking establishments, are the direct beneficiaries. NGOs and government institutions support implementation and monitoring. The composition of stakeholder groups has remained largely unchanged over time.
Expected changes in well-being	The project is expected to improve public health by reducing indoor air pollution (IAP), enhance economic conditions by promoting access to clean cooking solutions, and support job creation through project implementation and monitoring. It also contributes to environmental protection by eliminating the biomass demand for cooking, which helps reduce deforestation and forest degradation. Collectively, these outcomes contribute to poverty alleviation, women's empowerment, and long-term ecological sustainability.
Location of stakeholders	Stakeholders are primarily located within the project area, which includes areas where the LPG stoves are being distributed. The benefits of improved cookstoves are localized, with no expected impacts on communities or stakeholders outside the defined project boundary.
Location of resources	The project is implemented in various regions across Ghana where targeted stakeholders—primarily small-scale commercial kitchen operators—reside and operate. These stakeholders do not surrender or transfer ownership of any land or natural resources as part of the project. However, many beneficiaries traditionally rely on local forests and communal lands for the collection of firewood and charcoal used for cooking. These natural resources are typically accessed through customary rights within their communities. By introducing energy-efficient LPG stoves, the project reduces dependence on these customary biomass resources, thereby helping to alleviate pressure on forested areas and contributing to the sustainable management of community lands and natural ecosystems. While no direct access restrictions or changes in land tenure arise from project implementation, the intervention supports long-term conservation of the resources to which local stakeholders traditionally have access.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	21-May-2024
Stakeholder engagement process	The project engaged stakeholders through appropriate and inclusive methods, including phone calls, newspaper advertisements, public notices, emails, and online postings. These announcements were made well in advance of the Local Stakeholder Consultation (LSC) meeting, which was held on 21st May 2024, from 10:00 AM to 12:00 PM at Lancaster Hotel, Accra. The consultation was conducted in an appropriate language and encouraged participation of women through targeted outreach. The sessions featured presentations, project device demonstrations, and open discussions to ensure engagement from all participants. Key outcomes were documented through photos, videos, attendance records, and written feedback
Consultation outcome	The consultation engaged all relevant stakeholders in discussing the project's design, implementation, and potential impacts. The discussion covered potential risks, costs, and benefits, emphasizing the positive impact of clean fuel and technology for cooking on health and environment. The project complies with all relevant labor laws in the host country, ensuring fair wages, equal opportunities, worker protection and VCS validation and monitoring process. Free, Prior, and Informed Consent (FPIC) was respected, with stakeholders fully informed about the program, ensuring no infringement on property rights. The transfer of carbon rights to the PP was also communicated to stakeholders at the time of meeting.

	
Ongoing communication	The project maintains ongoing stakeholder engagement through various communication channels: • Continuous input / Grievance Expression process book is available at office of Envirofit in Ghana. • Email Communication – Stakeholders can reach PP via the official email for inquiries and feedback. • Follow-Up Visits – Local people employed as field staff, and resource persons also serve as medium to escalate grievances received from the project beneficiaries in their respective zones to PP. The staff conduct regular household visits to monitor progress, address concerns, and ensure proper cookstove usage. • Telephonic communication: Stakeholders can reach PP via the official customer care contact number for inquiries and feedback.
Stakeholder input	Stakeholder feedback was actively documented and addressed. A detailed Q&A was conducted during the consultation. Key queries included stove access, payment terms, training, partnerships, fuel types, quality assurance, and the role of carbon revenues. All input was acknowledged and answered. No changes to project design were deemed necessary, but the feedback reinforced the importance of affordability, awareness, and technical support for end users. Below are the queries raised by the attendees: Que.1 Is Envirofit open for partnerships? Ans.1 Yes, Envirofit is open to partnerships with NGO's, social service groups, and others to assist in bridging gaps in the supply chain, creating awareness and demand, and making the technology more accessible to the targeted users. Through these partnerships we can achieve greater reach and impact. Que.2 What end users is the project targeting? Ans.2 The project targets the commercial settings (i.e., restaurants'), institutions, and communities, etc.

Que.3 Will the stoves be distributed for free or will they be sold. If they are being sold, is payment required upfront, or can it be made in installments?

Ans.3 Yes, the ICS will be distributed to the beneficiaries at a subsidized cost, with the option to pay in installments over 6 – 12 months.

Que.4 How do customers get access to the LPG stoves?

Ans.4 There are specific criteria established by Envirofit for distribution of LPG stoves. LPG stoves are distributed to commercial settings and institutions that currently use traditional stoves with solid fuel. Envirofit staff visit potential customers to access their cooking style and cost by collecting end-user details. Then introduce the product to them and arrange for its delivery.

Que.5 Is any training provided to customers for the usage of LPG stoves?

Ans.5 Yes, the Envirofit team provides training to end users while distribution of LPG stoves. This training includes instructions for both owners and cooks on the proper usage and maintenance of the LPG stoves.

Que.6 Are the cylinders owned by the operators or they simply refilled for them?

Ans.6 This project is following the Cylinder Recirculation Method (CRM) where users do not own the gas cylinders they use. Instead, they exchange empty cylinders for filled ones.

Que.7 Do the people involved in the baseline study become potential customers or not?

Ans.7 Yes, Participants in baseline study are considered potentials customers and some of them were also present at the meeting.

Que.8 Who checks the quality of LPG burners being supplied?

Ans.8 The Ghana Standard Authority (GSA) conducts check on the quality of LPG burners and Cylinder.

Que.9 What is the role of the carbon credit revenue generated from the project?

Ans.9 The carbon credit revenue from the project will help overcome the primary barrier, i.e., price for end users, by providing discounts on the cost of the stove.

Que.10 What baseline fuels are being replaced?

Ans.10 End users using woodfuel and charcoal as their baseline fuel will be replaced by the project.

Que.11 Will annual KPTs be conducted to estimate the baseline and project fuel consumption?

Ans.11 Baseline and project KPTs are the monitoring parameters that will be conducted in accordance with applied methodology and emission reductions are calculated accordingly.

Que.12 Is the project also focusing on the households along with the commercial users?

Ans.12 The target beneficiaries of the project are the commercial and Institutional users. However, Envirofit is also planning for a similar project for households in Ghana.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project does not involve land acquisition, resettlement, or any restriction on access to land or natural resources. It solely involves the voluntary distribution of LPG cookstoves to commercial kitchens in Ghana. Consent was obtained during the stakeholder consultation held on 21 st May 2024, where information regarding the project objectives, benefits, risks, and implementation methods was shared in a transparent and appropriate manner. The meeting included participation from local community representatives, small commercial users, NGOs, and other key stakeholders. No Indigenous Peoples (IPs) or customary rights holders were displaced or negatively impacted by the project. No ongoing or unresolved conflicts were identified or reported during the stakeholder engagement process, and the project does not exacerbate or influence any existing disputes. Upon distribution, beneficiaries formally transfer their rights to carbon credit ownership through a signed agreement. This process ensures that Free, Prior, and Informed Consent (FPIC) is obtained by the implementation partner for carbon credit ownership.
Outcome of FPIC	The outcome of the FPIC process confirmed that stakeholders were fully informed of the project's design, benefits, and implications. The consultation included clear communication about voluntary participation, training, carbon transfer rights to PP and grievance redressal methods. Stakeholders expressed support for the project and its objectives. No objections were raised as the project does not infringe on any landownership, tenure rights, or traditional resource use. The project has not resulted in any forced physical or economic displacement, and all project LPGS distributions are made with the voluntary consent of recipients.

2.1.4 Grievance Redress Procedure

Development process	Grievances can be submitted via a call center, email, or through an expression book ensuring accessibility in local languages and cultural alignment. Every grievance is carefully recorded, reviewed, and resolved transparently within a reasonable timeframe. Stakeholders were free to voice their concerns via the Grievance Expression Book. During the Local Stakeholder Consultation (LSC), stakeholders expressed strong support for the project, highlighting its benefits in promoting clean cooking, reducing costs, and creating job opportunities, with no significant concerns raised.
Grievance redress procedure	<i>Describe the grievance redress procedures developed with stakeholders.</i>

2.1.5 Public Comments

Comments received	Actions taken
Refer section 2.1.2 above	No action is deemed required based on comments received.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Demonstrate that management teams have expertise or experience in implementing similar project activities and engaging communities. Where relevant experience is lacking, demonstrate how the project proponent has partnered with other organizations to support the project or have a recruitment strategy to fill the identified gaps.

2.2.2 Risk Assessment

Use the table below to describe the risk assessment and outcome of the potential risks to stakeholders and the environment. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write “No risk identified” in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing		
Risks to stakeholder participation		
Working conditions		
Safety of women and girls		
Safety of minority and marginalized groups, including children		
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)		

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Use the table below to identify and summarize the risks for rights related to labor and work. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write “No risk identified” in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified ¹²	Mitigation or preventative measure(s) taken
Discrimination		
Sexual harassment		
Equal pay for equal work		
Gender equity in labor and work		
Forced labor		
Child labor		
Human trafficking		

2.3.2 Human Rights

Use the table below to identify and summarize any risks related to recognizing, respecting, and promoting the protection of the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write “No risk identified” in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation or preventative measure(s) taken

2.3.3 Indigenous Peoples and Cultural Heritage

Use the table below to identify and summarize any risks related to recognizing, respecting, and promoting the protection of the rights of IPs, LCs, and customary rights holders, and to preserving and protecting cultural heritage as part of project activities. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write “No risk identified” in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation(s) or preventative measure taken

¹² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

2.3.4 Property Rights

Use the table below to identify and summarize any risks related to protecting and preserving the property rights of IPs, LCs, and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation or preventative measure(s) taken

2.3.5 Benefit Sharing

Where the project impacts property rights as described in Section 2.4.4 above, use the table below to describe the project's benefit sharing agreement.

Process used to design the benefit sharing plan	Describe the process used to develop the benefit-sharing agreement with the affected stakeholder groups.
Summary of the benefit sharing plan	Describe the benefit-sharing agreement. Where affected stakeholder groups wish to keep elements of the plan private, provide the full arrangement as a commercially sensitive document. The project proponent shall demonstrate that the community wishes to keep this information private.
Approval and dissemination of benefit sharing plan	Demonstrate that the benefit-sharing agreement was agreed up on by the affected stakeholder groups, and that the agreement was shared in a culturally appropriate manner. Demonstrate that the agreement is readily accessible should stakeholders wish to review the agreement.

2.4 Ecosystem Health

Use the table below to identify and summarize any risks related to impacts on biodiversity and ecosystems, soil degradation and soil erosion, and water consumption and stress. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems		
Soil degradation and soil erosion		
Water consumption and stress		

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☐ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	Demonstrate that the project will not adversely impact habitats and areas needed for habitat connectivity for rare, threatened, or endangered species.
Areas needed for habitat connectivity	Demonstrate that the project will not adversely impact areas needed for habitat connectivity.
...	...

Use the table below to identify and summarize any risks related to habitats for rare, threatened, and endangered species, and for areas for habitat connectivity. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species		
Areas for habitat connectivity		

2.4.2 Introduction of Species

Demonstrate, using the table below, that no invasive species will be used as part of project activities. Categorize each species as native, non-native, and indicate if the species is a monoculture. Where the species is non-native, include an explanation of possible adverse effects of its use and a description of how the project will mitigate such risks. For projects with no planting or species introduction, this section may be indicated as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

Where invasive species exist in the project area, list such species in the table below and describe the commensurate mitigation measure(s) in place to prevent the spread or continued existence of invasive species.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species

Use the table below to identify and summarize any risks related to invasive species. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species		

2.4.3 Ecosystem Conversion

ARR, ALM, WRC or ACoGS projects shall provide evidence that the project area was not cleared or drained of existing natural ecosystems, unless such clearing took place at least 10 years prior, or the dominant land cover was invasive.

Use the table below to identify and summarize any risks related to ecosystem conversion. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion		

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Project will involve PAIs (LPGS) that will apply the following methodology and tool, approved under VERRA:

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	Energy Efficiency and Fuel-Switch Measures in Cookstoves https://verra.org/wp-content/uploads/2024/10/VM0050-EE-and-Fuel-Switch-Measures-in-Cookstoves-v1.0.pdf	1.0

CDM TOOL	12	Project and leakage emissions from transportation of freight https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf	1.1
CDM TOOL	15	Upstream leakage emissions associated with fossil fuel use https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-15-v2.0.pdf	2.0
CDM TOOL	30	Calculation of the fraction of non-renewable biomass https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf	4.0
CDM TOOL	33	Default values for common parameters https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v1.pdf	1.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050	Energy Efficiency and Fuel-Switch Measures in Cookstoves	
VM0050	Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with: i. More efficient project devices that use the same fuel as in the baseline. ii. Efficient project devices fired by renewable biomass or bioethanol. iii. Efficient project devices fired by liquefied petroleum gas (LPG); or iv. Electric-powered project devices.	The project will replace non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with project devices fired by liquefied petroleum gas (LPG).
VM0050	Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following: i. Efficient project devices fired by renewable biomass or bioethanol ii. Efficient project devices fired by LPG; or iii. Electric-powered project devices.	The project will also replace solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with efficient project devices fired by liquefied petroleum gas (LPG).
VM0050	Project devices are used in households, communities, institutions, or small or medium enterprises	The project involves the distribution of energy-efficient LPG stoves in commercial kitchens operating within small and medium-sized enterprises (SMEs) in Ghana

	(SMEs) ⁴ collectively referred to in this methodology as the “target population.	hereby serving the "target population" as defined by the methodology.
Use of Renewable Biomass		
VM0050	Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; or b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16.	The applicability conditions listed under the “Use of Renewable mass” are not applicable to the project as the project does not involve the use of renewable biomass. Rather, it involves a fuel switch from non-renewable biomass (e.g., firewood, charcoal) or higher carbon intensive fossil fuel to Liquefied Petroleum Gas (LPG), a cleaner and more efficient cooking fuel.
VM0050	Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land areas from which the biomass residues originate.	
VM0050	Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.	
VM0050	The project does not use multiple types of renewable biomass as a fuel source but instead improves the efficiency of existing biomass use, reducing overall fuel consumption.	
VM0050	Where project activities introduce renewable biomass as charcoal, it is renewable charcoal produced by efficient charcoal production processes ⁶ (e.g., retort sedentary) kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	
Cookstove characteristics ¹³ and use		
VM0050	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved	Not applicable, as the condition pertains to cookstoves using biomass fuels (renewable

¹³ Only thermal efficiency and durability of cookstoves are evaluated.

	efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%	or non-renewable) but the project involves LPG as the fuel.
VM0050	Project devices using LPG or bioethanol are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 30%	The project involves the distribution of single-pot and/or multi-pot portable LPG stoves, designed for commercial kitchens. Based on performance specifications and testing data, these stoves demonstrate initial thermal efficiency greater than 30%.
VM0050	Electric project devices meet the maximum risk factor score of 15 on the Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% b) Induction stoves and other electric stoves: 70%	Not applicable, as the condition pertains exclusively to electric cooking devices, whereas the project involves the distribution of LPG stoves to target population.
VM0050	Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon intensive fossil fuel (demonstrated by the baseline survey, see Section 6.2) b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The project does not seek to issue any carbon credits for periods after 31 December 2045.	a) The baseline fuel used by targeted commercial kitchens shall be predominantly comprised of either non-renewable biomass or more carbon intensive fossil fuel. This shall be confirmed through the baseline survey conducted in line with the methodology requirements and the agreement with each beneficiary b) The project includes a monitoring system to track LPG consumption, either through metering at LPG filling stations or via verified fuel purchase records linked to each consumer. This provision enables accurate determination of actual LPG usage per project consumer, in compliance with the methodological requirement. c) PP confirms that carbon credits shall not be claimed for periods after 31 December 2045, in accordance with the upper temporal boundary defined in the methodology.
VM0050	Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less than 1% of the annual electricity of the decentralized renewable energy system.	Not applicable, as the condition pertains exclusively to electric cooking device, whereas the project involves the distribution of LPG stoves to target population.

	b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	
VM0050	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description.	The project proponent has designed incentive mechanisms to eliminate the use of non-renewable biomass or more carbon intensive fossil fuel in the baseline by distributing project LPGs at subsidized cost to beneficiaries. Additionally, awareness programs and community engagement initiatives are in place to ensure sustained adoption. These mechanisms are detailed in the project description.
VM0050	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	The project proponent ensures that once a project LPGs reaches the end of its technical life, it will either be replaced with a comparable or better project device or retrofitted to maintain its functionality and efficiency. Otherwise, emission reductions from such project devices will be claimed only till their rated technical lifetime.
Avoiding harm and double counting		
VM0050	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	PP has implemented a robust distribution and tracking mechanism to ensure unique identification of each project LPGS and to avoid double counting of emission reductions. At the time of distribution, each stove is marked with program logo/ unique alphanumeric ID that is physically labeled on the project device. Additionally, detailed end-user information—including name, contact details (if available), and end user location such as geographical coordinates/complete address information is recorded. This ensures traceability and transparency throughout the project device's lifecycle and prevents overlapping with any other mitigation initiatives.
VM0050	The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy	The project complies with all applicable national and local regulations governing the distribution and use of clean cooking technologies and associated fuels in Ghana. The project involves the distribution of energy-efficient LPG cookstoves to commercial kitchens, replacing traditional biomass-based cooking systems.

	services provided by the project must be documented.	A review of Ghana's regulatory framework indicates the following: • There are no mandatory regulations that restrict or prohibit the installation, commercialization, or distribution of LPG stoves in the country. • The Government of Ghana (GoG) actively supports clean cooking solutions, including LPG, through policies such as: ◦ The National LPG Promotion Policy (2017), which seeks to increase the adoption of LPG for cooking among households and commercial entities. ◦ The Ghana Sustainable Energy for All (SE4ALL) Action Plan, which prioritizes access to modern energy services, including clean cooking technologies. ◦ The Energy Commission Act, 1997 (Act 541), which empowers to promote energy efficiency and productive uses of electricity, natural gas and petroleum products.
VM0050	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	Not applicable. The project does not involve REDD+.
CDM Tool15	Upstream leakage emissions associated with fossil fuel use	
CDM Tool15	The tool is applicable to calculate leakage upstream emissions associated with fossil fuel use in either or both the baseline scenario and project.	The project involves a fuel switch from non-renewable biomass to LPG that is a fossil fuel. Therefore, the tool can be used for calculating upstream emissions associated with LPG use in the project scenario.
CDM Tool15	Upstream emissions associated with fossil fuel use are: (a) Fugitive emissions of CH ₄ and CO ₂ (e.g., venting, flaring, leaks); (b) CO ₂	The project's LPG fuel has upstream emissions arising from production, transportation, and distribution. These

	emissions from combustion of fossil fuels; (c) CO ₂ emissions associated with electricity consumption.	sources are covered by the tool, making it suitable for the project.
CDM Tool15	Other GHG emissions sources, such as those from construction of equipment, are relatively small and not considered.	The project does not include emissions from stove manufacturing or infrastructure development, in line with the tool's scope.
CDM Tool15	The tool has two options: Option (A): Simple approach using default emission factors; Option (B): Detailed approach based on upstream emissions stages.	For calculating the upstream leakage emissions associated with fossil fuel use one of the two options shall be used. Also, as per para 20 of the tool, the approach may be changed during the crediting period, however, a combination of Options (A) and (B) shall not be used in the same monitoring period. Thus, the final approach adopted in a particular monitoring period shall be documented in the monitoring report.
CDM Tool15	Applicable fossil fuels include those based on natural gas, oil, or coal:	The project involves a fuel switch from non-renewable biomass or carbon-intensive fossil fuels to LPG, a cleaner and more efficient energy source for cooking. LPG is obtained through two primary processes: • Separated from natural gas during its processing. • Produced during the refining of crude oil. Thus, the applicable fossil fuel is natural gas.
	Methodologies which refer to this tool should state:	
CDM Tool15	a) Methodologies using this tool should state the fossil fuel type(s) for which upstream emissions are determined.	The project involves a fuel switch from non-renewable biomass or carbon-intensive fossil fuels to LPG, a cleaner and more efficient energy source for cooking, on page 23 of the applied methodology following has been clearly stated: <i>“Projects using LPG must apply CDM TOOL15 Upstream Leakage Emissions”</i>
CDM Tool15	b) Procedures to determine the amount of fossil fuel used (on a net calorific value basis).	Procedures to determine the amount of LPG used in the project scenario has been provided in section 8.2.1.1 of the applied methodology and using equation (4) it can be converted into TJ using standard NCVs.
CDM Tool15	c) If net leakage values are calculated as negative, they are to be considered zero (unless otherwise permitted).	In cases where the project fuel replaces non-renewable biomass in the baseline scenario, upstream leakage emissions will be positive, as the baseline fuel is not a fossil fuel and therefore its upstream emissions are considered zero.

		In accordance with the tool's guidance, if any net upstream emissions are calculated as negative, they shall be conservatively set to zero.
CDM Tool30	Calculation of the fraction of non-renewable biomass	
CDM Tool30	This tool may be used by: a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or b) project participants to calculate project- or PoA-specific f_{NRB} values.	Option (b) shall be applied to calculate the project specific f_{NRB} value.
CDM Tool30	Project participants and DNAs shall identify and clearly delineate the applicable area for which f_{NRB} is determined. For project- or PoA-specific f_{NRB} values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project and justify the selection of the area in CDM project design documents. For region- or country-specific f_{NRB} values, DNAs shall specify the applicable area.	The PP has chosen host country Ghana as region for the project.
CDM Tool30	Project participants and DNAs may choose between the following two options to determine the value of f_{NRB}: (a) Use the default value as provided in TOOL33; or (b) Calculate f_{NRB} by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and requirements in the paragraphs below. The project participants shall compare and analyse the calculated values against the values for f_{NRB} reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least:	For determining the value of f_{NRB} , one of the two options shall be used. For detail refer f_{NRB} calculator.

	(i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.	
CDM Tool30	If the f_{NRB} value is estimated at the national level, as a cross check, project proponent shall compare the value of estimated NRB with the product of: i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr). If the estimated NRB value is more than 10% above the value calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB	For details refer to the f_{NRB} calculator.

3.3 Project Boundary

The boundary for the Project in terms of a geographical area is defined as the territorial boundary of Ghana. The boundary for each PAI corresponds to the physical location where the distributed LPGS units are used within Ghana.

Source		Gas	Included?	Justification/Explanation
Baseline	Meeting Thermal energy needs from use of biomass / carbon intensive fossil fuels	CO ₂	Yes	Major source
		CH ₄	Yes	Significant source
		N ₂ O	Yes	Significant source
		Other	No	Not applicable
	Production of charcoal fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Significant source
		N ₂ O	Yes	Significant source
		Other	No	Not applicable
Project	Meeting Thermal energy needs from use of Project LPGS	CO ₂	Yes	Major source
		CH ₄	Yes	Significant source
		N ₂ O	Yes	Significant source
		Other	No	Not applicable
		CO ₂	Yes	Major source
		CH ₄	Yes	Significant source

Source	Gas	Included?	Justification/Explanation
Upstream emissions from LPG	N ₂ O	Yes	Significant source
	Other	No	Not applicable

Provide a diagram or map of the project boundary, showing clearly the physical locations of the various installations or management activities taking place as part of the project activity based on the description provided in Section 1.12 (Description of the Project Activity) above.

For non-AFOLU projects, include in the diagram the equipment, systems and flows of mass and energy. Include the GHG emission sources identified in the project boundary.

For AFOLU projects, include in the diagram or map the locations of where the various measures are taking place, any reference areas and leakage belts.

3.4 Baseline Scenario

Identify and justify the baseline scenario, in accordance with the procedure set out in the applied methodology and any relevant tools. Where the procedure in the applied methodology involves several steps, describe how each step is applied and clearly document the outcome of each step.

Explain and justify key assumptions, rationale, and methodological choices. Provide all relevant references.

3.5 Additionality

Demonstrate and assess the additionality of the project, in accordance with the applied methodology and any relevant tools, taking into account the following additionality methods:

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

3.5.2 Additionality Methods

- Where a project method is applied to demonstrate additionality and the procedure in the applied methodology or tool involves several steps, describe how each step is applied and clearly document the outcome of each step. Indicate clearly the method selected to demonstrate additionality (e.g., investment analysis or barrier analysis in the case of the CDM Tool for the demonstration and assessment of additionality). Where barrier analysis, or equivalent, is used to demonstrate additionality, only include the most relevant barriers. Justify the credibility of the barriers with key facts and/or assumptions and the rationale. Provide all relevant references.

- Where a performance method is applied to demonstrate additionality, demonstrate that performance can be achieved to a level at least equivalent to the performance benchmark metric.
- Where the methodology applies an activity method for the demonstration of additionality, include a statement that notes that conformance with the positive list is demonstrated in the Applicability of Methodology section above.
- Provide sufficient information (including all relevant data and parameters, with sources) so that a reader can reproduce the additionality analysis and obtain the same results.

3.6 Methodology Deviations

Describe and justify any methodology deviations applied, including any previous deviations. Include evidence to demonstrate the following:

- The deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.
- The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Describe the procedure for quantification of baseline emissions and/or carbon stock changes in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

4.2 Project Emissions

Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

4.3 Leakage Emissions

Describe the procedure for quantification of leakage emissions in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.

For data and parameters monitored, use the estimated data/parameter values provided in Section 5.2 below. Document how each equation is applied in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCU for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 10,000	Example: 10,000	Example: 20,000
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						
Total						

For projects required to assess permanence risk:

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	Example: 20%
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Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No If no, provide justification.

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO₂e removals and 80,000 tCO₂e reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
DD- MMM- YYYY to 31-Dec- YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000	Example: 16,000
01-Jan- YYYY to 31-Dec- YYYY							
01-Jan- YYYY to DD- MMM- YYYY							
Total							

5 MONITORING

5.1 Data and Parameters Available at Validation

Complete the table below for all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above. Data and

parameters to be monitored during the operation of the project are included in Section 5.2 (Data and Parameters Monitored) below.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Value applied	Provide the value applied
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.
Purpose of data	Indicate one of the following: - Determination of baseline scenario (AFOLU projects only) - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Comments	Provide any additional comments

5.2 Data and Parameters Monitored

Complete the table below for all data and parameters that will be monitored during the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Description of measurement methods and procedures to be applied	Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).

Frequency of monitoring/recording	<i>Specify measurement and recording frequency</i>
Value applied	<i>Provide an estimated value for the data/parameter</i>
Monitoring equipment	<i>Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.</i>
QA/QC procedures to be applied	<i>Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.</i>
Purpose of data	<i>Indicate one of the following:</i> • <i>Calculation of baseline emissions</i> • <i>Calculation of project emissions</i> • <i>Calculation of leakage</i>
Calculation method	<i>Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.</i>
Comments	<i>Provide any additional comments</i>

5.3 Monitoring Plan

Describe the process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters set out in Section 5.2 (Data and Parameters Monitored) above.

Include details on the following:

- *The methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the procedures for calibrating monitoring equipment.*
- *The organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.*
- *The procedures for internal auditing and QA/QC.*
- *The procedures for handling non-conformances with the validated monitoring plan.*
- *Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.*

Where appropriate, include line diagrams to display the GHG data collection and management system.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.